

# Enhancing low-carbon tourism based on local resources and carbon footprint assessment of community tourism routes in Ban Rak Thai, Mae Hong Son



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**Abstract** This study explores the development of low-carbon tourism (LCT) using local resources and evaluates the carbon footprint of community-based tourism routes in Ban Rak Thai, Mae Hong Son Province. Employing a mixed-methods approach, the research integrates quantitative data from 385 Thai tourists with qualitative insights from 35 key informants through surveys, interviews, and participatory community engagement. The analysis of tourism potential across eight dimensions highlights the area's rich natural landscape, distinctive Yunnanese-Chinese culture, sustainable agricultural practices, and strong community involvement. However, challenges remain, particularly in waste management, renewable energy usage, and eco-friendly transportation systems. Tourist behavior analysis indicates a strong willingness to engage in environmentally responsible travel, with a mean score of 4.39 out of 5. Two prototype one-day tourism routes were developed and tested with actual tourists to assess their carbon footprint. The results show that most activities, including hiking, tea-making workshops, and cycling, generated low or negligible greenhouse gas (GHG) emissions. Emissions were primarily linked to food consumption and fossil-fuel-based transportation. The study affirms the feasibility of LCT as a sustainable development model, aligning with national and global environmental goals. It also demonstrates that integrating scientific carbon footprint analysis with community-based tourism planning can lead to effective, localized strategies for sustainable tourism. Recommendations include promoting clean energy in local accommodations, improving waste infrastructure, and developing green transportation options. The findings offer a replicable framework for low-carbon rural tourism that is applicable to other destinations in Thailand and similar contexts worldwide. This research contributes to the broader discourse on sustainable tourism and climate change mitigation through community empowerment and informed planning.

**Keywords:** low-carbon tourism, carbon footprint, Ban Rak Thai, community participation, sustainable tourism

## 1. Introduction

Tourism has long been recognized as a key economic driver in Thailand, contributing significantly to national income and employment. In 2023, international tourism generated an estimated 1.9 trillion baht in revenue from over 39.92 million inbound visitors (Ministry of Tourism and Sports, 2025). Beyond its economic value, international tourism also contributes positively to Thailand's foreign exchange reserves. However, the COVID-19 pandemic severely disrupted global travel, resulting in unprecedented setbacks for the tourism sector. Although recovery was evident in 2023—reaching 71% of prepandemic levels and projected to rise to 89% in 2024 (approximately 35.5 million tourists)—Thailand's recovery remains slower than that of other major global destinations (United Nations World Tourism Organization, 2024).

In addition to economic concerns, growing attention has been given to the environmental sustainability of tourism. The overexploitation of natural resources—including finite assets such as minerals and fossil fuels—and renewable biological resources such as flora, fauna, and microorganisms—poses long-term environmental challenges. These biological resources are essential to Thailand's biodiversity, which serves as a key comparative advantage, particularly compared with many technologically advanced Western nations that possess less natural diversity (Sloan et al., 2013).

One of the most critical environmental issues associated with tourism is greenhouse gas (GHG) emissions, which significantly contribute to global warming (IPCC, 2023). Tourism-related activities—including transportation, accommodations, and consumption—are all linked to substantial carbon footprints. In response, the concept of low-carbon tourism (LCT) has emerged as a strategic approach to sustainable tourism development. LCT focuses on reducing GHG emissions throughout the tourism value chain by encouraging environmentally responsible travel behavior and the adoption of green technologies



(Gössling et al., 2018; Becken & Patterson, 2006). The successful implementation of LCT relies on community participation, strategic planning, and strong policy support (Sun et al., 2022).

A promising location for low-carbon and ecotourism initiatives is Ban Rak Thai, which is located in Mok Cham Pae Subdistrict, Mae Hong Son Province. Although villages are well known as cultural tourism destinations, they face challenges such as overcrowding during holidays, short visitor stays, and a limited variety of tourism activities and routes. These issues restrict both economic retention and the sustainable use of local resources (Matichon Online, 2023). Furthermore, existing tourism practices lack carbon monitoring mechanisms and participatory frameworks for local resource management. Despite receiving over 65,000 visitors annually (Limpanitgul et al., 2024), Ban Rak Thai lacks comprehensive data on local resource capacity, tourist behavior, and tourism-related carbon emissions. Moreover, there is no standardized method for measuring or mitigating the carbon footprint of tourism activities at the community level.

This study seeks to address these gaps by assessing the carbon emissions generated by tourism in Ban Rak Thai and proposing low-carbon tourism routes. The objective is to establish a framework for sustainable tourism resource management, reduce environmental impact, and enhance the community's capacity to deliver authentic and environmentally responsible tourism experiences.

## 2. Research Objectives

The objectives of this research are threefold: (1) to analyze the potential of local resource bases and the needs of the community regarding the development of low-carbon tourism in Ban Rak Thai, Mae Hong Son Province; (2) to examine the behaviors and preferences of tourists toward low-carbon tourism models in Ban Rak Thai; and (3) to assess the carbon footprint of prototype tourism routes developed through participatory processes involving the Ban Rak Thai community.

## 3. Research Methodology

This study employed a mixed-methods research design, integrating both quantitative and qualitative approaches.

### 3.1. Population and Key Informants

The target population consisted of 385 Thai tourists who had visited the Ban Rak Thai community in Mae Hong Son Province. The sample size was calculated via Cochran's formula (Cochran, 1963), with a 95% confidence level and a margin of error below 5%. A simple random sampling technique was used to ensure representativeness and minimize selection bias (Etikan & Bala, 2017).

The key informants were selected through purposive sampling, which focused on individuals with relevant knowledge and experience related to low-carbon tourism within the community. An initial list of potential participants was created, and additional informants were recruited via the snowball sampling method on the basis of referrals from earlier participants (Noy, 2008). The selection followed qualitative sampling principles outlined by Nastasi and Schensul (2005) and Miles and Huberman (1994). Criteria for data sufficiency and saturation were also applied (Onwuegbuzie & Leech, 2007; Flick, 1998), with saturation deemed reached when no new information emerged after more than five informants were interviewed. In total, 35 key informants were engaged, comprising 10 government agency representatives, 10 local entrepreneurs, 5 Thai community council members, and 10 local residents, in accordance with the guidelines of McMillan and Schumacher (1997).

### 3.2. Research instruments

A structured questionnaire based on a five-point Likert scale was developed to assess tourist behavior and preferences. The instrument demonstrated high internal consistency, with a Cronbach's alpha coefficient of 0.89, indicating acceptable reliability for social science research (Tavakol & Dennick, 2011).

Semistructured interviews were also conducted to examine the community's capacity for low-carbon tourism development. These interviews explored local tourism resources, strengths, weaknesses, opportunities, and threats (SWOT analysis). The analysis was guided by the tourism potential assessment framework proposed by Singsaktrakul and Sermkarndee (2013), which evaluates eight key dimensions: Tourism Resources, Accessibility, Infrastructure and Facilities, Tourism Activities, Management and Administration, Community Participation, Marketing and Promotion, and Environmental and Social Impacts. This framework aligns with the multidimensional tourism planning models found in the sustainable tourism literature (Crouch & Ritchie, 1999; Bramwell & Lane, 2011).

### 3.3. Data collection

Primary data were collected from March to December 2024 through questionnaires, field surveys, participant observations, in-depth interviews, and focus group discussions. The secondary data included government reports, ecotourism publications, peer-reviewed journal articles indexed in Scopus, and academic research from the Thai-Journal Citation Index Centre. The use of multiple sources follows established triangulation practices to enhance data validity (Denzin, 2012).

### 3.4. Data Validation

To ensure credibility, researcher checks were conducted during and after data collection. Each key informant was given the opportunity to review their responses, and additional clarifications were sought when necessary. This process aimed to ensure that the recorded data accurately reflected the participants' intentions, thereby enhancing the trustworthiness and authenticity of the findings (Lincoln & Guba, 1985).

### 3.5. Data analysis

The quantitative data, including percentages, means, frequencies, and standard deviations, were analyzed via SPSS to generate descriptive statistics. Qualitative data were analyzed via content analysis and thematically synthesized on the basis of Colaizzi's (1978) phenomenological approach. The findings were validated through triangulation, as recommended by Cohen and Manion (1994) and Miles and Huberman (1994), ensuring rigor through cross-verification from multiple data sources and methods.

### 3.6. GHG Emissions Assessment

Greenhouse gas (GHG) emissions were assessed via the following equation:

$$\text{GHG emissions} = \text{Activity Data} \times \text{Emission Factor} \quad (1)$$

The emission factor represents the quantity of greenhouse gases emitted per unit of activity. Emission factors were adapted to reflect tourism-specific activities in Thailand, in line with the IPCC Guidelines (IPCC, 2006) and the Product Category Rules (PCR) for Tourism Services in Thailand (Thaicarbonlabel, 2022). This methodological approach is widely applied in environmental impact assessments related to tourism and carbon accounting (Gössling et al., 2005; Peeters & Dubois, 2010).

## 4. Results

The results of the study revealed the potential of local resource bases and the community's readiness for developing low-carbon tourism in Ban Rak Thai, Mae Hong Son Province.

### 4.1. SWOT Analysis of Low-Carbon Tourism Development in Ban Rak Thai, Mae Hong Son Province

Based on the study results, the researcher summarized the SWOT analysis in the following key points. Strengths (S): The lake and pine forest landscape, combined with the cool climate throughout the year, support low-energy activities such as walking, cycling, and low-impact boating. The Yunnan cultural way of life and tea products represent cultural capital that can be further developed. In addition, the community homestay system and local conservation groups help reduce management conflicts and effectively support green initiatives. Weaknesses (W): Access to the area relies heavily on private cars, while accommodation and restaurants still depend on fossil-based electricity. Waste management and single-use plastic reduction systems are not yet fully integrated, and carbon-reduction measures in the tourism sector remain underdeveloped. Opportunities (O): The growing demand for sustainable tourism, together with the national BCG policy and the advancement of EV and solar technologies, creates favorable conditions for investment. There are opportunities to develop "slow tourism," "creative tourism," and "agri-tourism" packages based on tea culture and Yunnan traditions, as well as to introduce clean-energy transportation within the area. Threats (T): A long rainy season and severe cold cause fluctuations in demand. There is also the risk of overtourism resulting from excessive promotion, uneven levels of acceptance and readiness among operators to invest in green technologies, and competition from nearby destinations. A detailed overview of these findings is presented in Table 1.

From the researcher's perspective, low-carbon tourism can be described as a spatial development framework that emphasizes reducing greenhouse gas emissions while simultaneously preserving natural and cultural capital and enhancing the quality of life in local communities. This study synthesizes both the potential and limitations of Ban Rak Thai through a SWOT framework and proposes a practical carbon footprint assessment model at the destination level. Such a model can serve as a foundation for designing tourism policies and products that facilitate the transition toward low-carbon development.

*4.2. The analysis of the potential of local resource bases and the community's readiness for developing low-carbon tourism in Ban Rak Thai, Mae Hong Son Province—based on the eight criteria outlined by Singsaktrakul and Sermkarndee (2013) and conducted through a participatory community process—revealed the following:*

#### 4.2.1. Tourism Resources

Ban Rak Thai is located near the Thai–Myanmar border in Village No. 6, Mok Cham Pae Subdistrict, Mueang District, Mae Hong Son Province. Its year-round cool climate and diverse natural and cultural tourism resources make it a distinctive destination. Key attractions include:

(1) *Khuean Nai Mok* (the Misty Reservoir), the most popular site in the village—a scenic lake surrounded by mountains in the heart of the community—is ideal for viewing sunrises and sunsets;

(2) *Ban Rak Thai tea plantations*, such as Rai Cha View Doi Rak Thai, Rai Cha Kung Fu Camp, Rai Cha Chen Feng, Rai Cha Rewine, and Rai Cha Cha Sa Rak Thai, which serve as agrotourism sites offering tea production tours, tasting experiences, and tea as souvenirs. Many guesthouses and shops in the village are built in the traditional Yunnanese clay-house architectural style, reflecting the community's identity—an initiative jointly agreed upon by the village administrative committee to preserve the uniqueness of Yunnan-Chinese heritage;

(3) *Wat Pa Rak Thai*, an important religious site and spiritual center for both locals and visitors. The temple showcases a blend of Thai and Chinese Buddhist art, symbolizing the community's cultural roots. Surrounding nature, including large trees and pine forests, its peaceful setting is ideal for meditation and spiritual retreats. Visitors can enjoy walking around the temple and experiencing the pristine natural beauty of Ban Rak Thai;

(4) Cultural tourism based on the Yunnan-Chinese way of life, with traditional customs, cuisine, and language passed down by Yunnanese migrants who settled in the area, giving Ban Rak Thai a unique identity distinct from that of other tourist destinations.

#### 4.2.2. Accessibility

Accessibility to Ban Rak Thai remains somewhat challenging due to its mountainous terrain. The village is situated at an elevation of more than 1,776 m above sea level, although road conditions have improved in recent years. The primary route is Highway No. 1095 (Mae Hong Son–Pai), from which travelers must turn onto a winding road to reach Ban Rak Thai. From the northern approach, a left turn around kilometer 8 leads toward Kung Mai Sak Village, along the same route that passes Phu Klon Country Club, Pha Suea Waterfall, and Pang Ung Village. The steep and winding roads require skilled driving, and vehicles must be refueled in advance, as there are no petrol stations on the mountain. The distance from Mae Hong Son town to Ban Rak Thai is approximately 44 kilometers. From Pai District, travelers pass through Pang Mapha District and continue past Tham Pla (Fish Cave). Approximately one kilometer beyond the cave, a right turn leads to Mok Cham Pae Village. According to the results of these interviews, public transportation remains limited, so most visitors prefer to travel via private cars or rental vehicles. However, during the rainy season, travel can be difficult because of slippery roads and dense fog in some areas.

#### 4.2.3. Infrastructure and Facilities

The infrastructure and facilities in Ban Rak Thai have significantly improved in recent years. The village offers a variety of accommodations, ranging from homestays and resorts to Yunnan-style lodging, such as the popular Lee Wine Rak Thai Resort. In terms of dining, there are several Yunnan-Chinese restaurants, including the well-known Ban Rak Thai Yunnan Pork Leg Restaurant, which highlights the village. Electricity and water supply systems are generally sufficient, although water shortages may occur during the dry season. Internet and mobile phone coverage are available, but some mobile networks have weak signals, especially in areas farther from the village center. Additionally, according to local residents, some outlying areas beyond the main tourist zones still lack access to electricity.

#### 4.2.4. Tourism Activities

The main tourism activities in Ban Rak Thai focus on cultural and nature-based tourism. Key activities include:

(1) At *Khuean Nai Mok*, visitors can enjoy boating in traditional Chinese style to admire the natural scenery. The boats are powered by solar-generated electricity used for lighting, and visitors can also sip Chinese tea while on board.

(2) Within *Ban Rak Thai* village, popular activities include participating in morning alms, taking cycling tours to explore the village and ride around the reservoir, and joining community tours via three-wheeled motorcycles covering a distance of 3.5 kilometers. Additional experiences include coffee workshops, green tea royal jelly workshops, and the purchase of dried oolong tea as souvenirs.

(3) At *Wat Pa Rak Thai*, visitors can practice *Vipassana* meditation, pay respect to sacred spaces without burning incense, and join guided walking tours through the forest garden.

#### 4.2.5. Management and Administration

Tourism management in Ban Rak Thai is generally good. A community tourism group operating under the Ban Rak Thai community administrative committee was established to care for visitors and preserve their local identity. However, some areas still require improvement. There is a lack of strict measures to control the number of tourists during the high season, which may negatively impact natural resources. Additionally, there is currently no systematic fee collection in place to support the development of public infrastructure.

#### 4.2.6. Community Participation

Community participation in the tourism industry in Ban Rak Thai is high, as tourism income is considered the main source of livelihood for local entrepreneurs. Nearly all the accommodation owners, restaurant operators, and tea producers are local residents. Community members have formed groups to develop tourist attractions and preserve cultural heritage. However, some members of the younger generation show limited interest in tourism-related work, resulting in reduced youth involvement in certain activities.

**Table 1** SWOT analysis of low-carbon tourism development.

| Strengths                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Weaknesses                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>- Ban Rak Thai features a natural lake surrounded by mountains and pine forests, which support environmentally friendly activities such as hiking, cycling, and low-energy boating.</li> <li>- The area enjoys cool weather throughout the year, which is conducive to low-carbon tourism, such as using bicycles or walking for sightseeing instead of pollution-emitting vehicles.</li> <li>- Yunnan-Chinese cuisine and culture—such as oolong tea, Yunnan-style pork leg, and steamed buns—attract tourists seeking traditional lifestyle experiences.</li> <li>- The community's nature-dependent way of life, including tea cultivation and organic farming, can be developed into low-carbon agritourism.</li> <li>- Ban Rak Thai has a homestay system operated by local residents, which helps promote low-carbon tourism through the use of local materials and waste reduction in accommodations.</li> <li>- Groups of villagers collaboratively manage conservation-based tourism, facilitating the implementation of low-carbon policies.</li> <li>- Traditional tea production in the community requires low energy and can be promoted as a low-carbon tourism product.</li> <li>- Souvenirs made from bamboo, bulrush, and ceramics are more sustainable than plastic alternatives.</li> </ul> | <ul style="list-style-type: none"> <li>- Ban Rak Thai is located in a remote area, and traveling to the village requires private vehicles or rental cars, as there is currently no environmentally friendly public transportation system, such as electric buses or public bicycles.</li> <li>- Renewable energy systems, such as solar panels and wind power, have not yet been fully developed.</li> <li>- Accommodations and restaurants still rely more on electricity generated from fossil fuels than on renewable energy sources.</li> <li>- Environmentally friendly waste management systems—such as waste separation, recycling, and the reduction of single-use plastics—are lacking.</li> <li>- There are still no measures in place to encourage accommodations and restaurants to reduce greenhouse gas emissions, such as by using sustainable materials or lowering electricity consumption.</li> </ul> |
| Opportunities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Threats                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <ul style="list-style-type: none"> <li>- Modern tourists place increasing importance on sustainable travel, presenting an opportunity for Ban Rak Thai to develop low-carbon tourism packages, such as conservation trekking or stays in homestays powered by renewable energy.</li> <li>- Government policies promoting low-carbon tourism, such as the BCG Model (Bio-Circular-Green Economy), can serve as a framework for developing context-appropriate projects.</li> <li>- There is an opportunity to introduce electric vehicles for transporting tourists, helping reduce travel-related pollution.</li> <li>- Solar power systems can be installed in homestays and restaurants to support clean energy use.</li> <li>- "Slow travel" packages can be developed, emphasizing walking, cycling, and participation in community-based activities such as tea cultivation or Yunnan cooking classes.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>- The long rainy season and extremely cold weather during certain times of the year may lead to a decline in tourist numbers, negatively affecting the local economy.</li> <li>- Overpromotion of tourism without proper regulation could turn Ban Rak Thai into a mass tourism destination, potentially leading to pollution and waste issues.</li> <li>- Nearby tourist destinations are becoming increasingly popular among travelers seeking nature-based and cultural experiences.</li> <li>- Although the community embraces sustainable practices, some business operators may still be unprepared or unwilling to invest in green technology.</li> <li>- Some tourist groups still place little importance on low-carbon tourism, continuing to use private vehicles and consume single-use plastic products.</li> </ul>                                                 |

#### 4.2.7. Marketing and Promotion

The marketing efforts in Ban Rak Thai have relied primarily on social media and word-of-mouth. Many tourists learn about the village through reviews on platforms such as Facebook, Instagram, YouTube, and TikTok. Promotions for accommodations and restaurants are also conducted through online travel platforms such as Agoda and Booking.com. However, there is still a lack of large-scale promotional campaigns from government agencies, which limits marketing outreach, primarily to tourists who are already familiar with the area.

#### 4.2.8. Environmental and Social Impacts

Tourism in Ban Rak Thai has generated both positive and negative impacts on the area. (1) On the positive side, tourism has become a source of income for the community, helped preserve Yunnan-Chinese culture, and contributed to the development of local infrastructure. (2) On the negative side, the growing number of tourists—especially during the New Year and long holiday periods—has led to increasing waste management challenges. Pollution has also resulted from business expansion, including the construction of new accommodations, an increase in the number of boat operators on the lake, and overcrowding during peak periods, all of which have contributed to the degradation of natural resources. The community currently lacks an internal waste disposal system and relies on transporting waste to Mueang District in Mae Hong Son Province.

#### 4.3. Analysis of Tourist Needs

A survey of 385 tourist respondents revealed that the majority were male, totaling 205 individuals (53.25%), followed by 145 (37.66%) female respondents and 35 (9.09%) LGBTQ+ individuals.

In terms of the age distribution in 2025, most respondents were Millennials (Gen Y), aged 28–43 years, accounting for 155 individuals (40.26%). This was followed by Gen X (44–59 years), with 101 respondents (26.23%); Baby Boomers (60 years and above), with 85 respondents (22.08%); and Gen Z (12–27 years), with 44 respondents (11.43%).

With respect to education level, the majority of respondents held a bachelor's degree (210 individuals or 54.55%), followed by those with education above the bachelor's level (154 individuals or 40.00%) and those with education below the bachelor's level (21 individuals or 5.45%).

In terms of occupation, most respondents were private company employees (135 individuals or 35.06%), followed by civil servants or state enterprise employees (112 individuals or 29.09%), entrepreneurs or business owners (109 individuals or 28.31%), and students (29 individuals or 7.53%). To further understand the preferences and travel behavior of these respondents, their needs were analyzed, as shown in Table 2.

#### 4.4. Results of Carbon Footprint Assessment for Two Prototype Tourism Routes Developed through a Participatory Community Process in Ban Rak Thai, Mae Hong Son Province

The study of the Ban Rak Thai community in Mae Hong Son Province is significant because of its popularity as a tourist destination and its contribution to greenhouse gas emissions, particularly from high-demand tourist activities. The researcher utilized data from the assessment of tourism resource potential and tourist needs related to low-carbon travel in Ban Rak Thai to design two prototype one-day trip routes. These routes were tested by a group of sample tourists, and their carbon footprints were evaluated accordingly, as illustrated in Figure 1.

**Table 2** Tourist travel needs and behavior.

| Tourist Need Statement                                                                                                                                                                                                                           | Mean ( $\bar{x}$ ) | S.D. | Level of Need |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------|---------------|
| 1. Do you desire to travel to destinations with a unique local identity, such as those featuring historical, cultural, and traditional elements that have been developed in an environmentally friendly manner?                                  | 4.31               | 0.67 | High          |
| 2. Do you wish to learn about local ways of life or cultures different from your own, particularly through approaches that promote carbon footprint reduction (e.g., using natural materials or participating in conservation-based activities)? | 4.30               | 0.66 | High          |
| 3. Do you value destinations known for products and souvenirs made from natural materials or created through environmentally friendly processes?                                                                                                 | 4.23               | 0.71 | High          |
| 4. Do you prefer to travel to areas with diverse tourism resources that have been developed to reduce environmental impact (e.g., nature trails or the use of clean energy at tourist sites)?                                                    | 4.51               | 0.63 | Very High     |
| 5. Do you consider it important to access tourist destinations via low-carbon transportation options (e.g., electric trams, rental bicycles, or walking)?                                                                                        | 4.35               | 0.69 | High          |
| 6. Do you prioritize travel to environmentally friendly destinations that implement measures to reduce greenhouse gas emissions?                                                                                                                 | 4.42               | 0.64 | High          |
| Average                                                                                                                                                                                                                                          | 4.35               | 0.67 | High          |

In addition to travel needs, the survey also examined expectations regarding accommodation and community services, summarized in Table 3.

**Table 3** Accommodation and services in the tourist community.

| Tourist Need Statement                                                                                                                                                             | Mean ( $\bar{x}$ ) | S.D. | Level of Need |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------|---------------|
| 1. Do you want the tourist community to be friendly and welcoming to visitors, and to provide information about reducing carbon footprints in tourism?                             | 4.43               | 0.65 | High          |
| 2. Do you require accommodations or hotels that are comfortable, meet safety standards, and emphasize green tourism concepts?                                                      | 4.45               | 0.67 | High          |
| 3. Do you prefer to stay in accommodations that implement energy conservation measures, such as the use of solar power, plastic waste reduction, and the use of natural materials? | 4.41               | 0.69 | High          |
| 4. Do you want accommodations to be conveniently located near tourist attractions and well connected to low-emission public transportation systems?                                | 4.44               | 0.66 | High          |
| 5. Do you want the tourist community to be friendly and welcoming to visitors, and to provide information about reducing carbon footprints in tourism?                             | 4.43               | 0.65 | High          |
| 6. Do you require accommodations or hotels that are comfortable, meet safety standards, and emphasize green tourism concepts?                                                      | 4.45               | 0.67 | High          |
| Average                                                                                                                                                                            | 4.43               | 0.67 | High          |

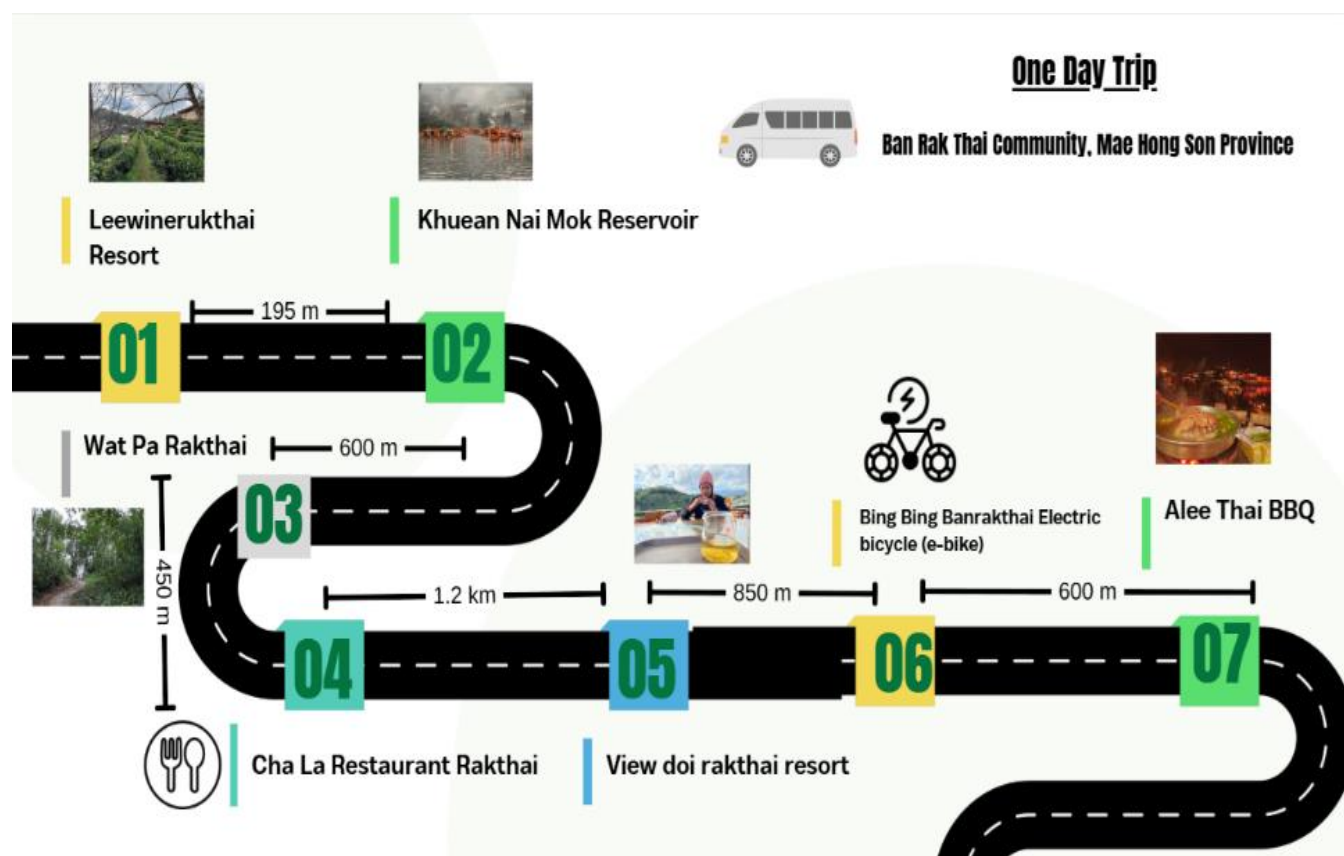
Finally, respondents' needs concerning quality and environmental management in tourism are presented in Table 4.

**Table 4** Tourist Needs for Quality and Environmental Management in Tourism.

| Tourist Need Statement                                                                                                                                                                                | Mean ( $\bar{x}$ ) | S.D. | Level of Need |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|------|---------------|
| 1. Do you value the cost-effectiveness of tourism products and services that align with sustainability principles and environmental friendliness?                                                     | 4.40               | 0.68 | High          |
| 2. Do you prefer tourism service processes that are convenient, easy to understand, and designed to minimize the use of paper or unnecessary documentation?                                           | 4.41               | 0.66 | High          |
| 3. Do you agree that tourism-related rules and regulations should promote environmental management and the reduction of carbon footprints?                                                            | 4.38               | 0.66 | High          |
| 4. Are you interested in learning about and experiencing ecosystems at tourist destinations, along with conservation practices that incorporate carbon footprint reduction?                           | 4.43               | 0.64 | High          |
| 5. Do you prefer to visit destinations managed according to sustainable tourism practices, such as waste separation, clean energy usage, or efficient water resource management?                      | 4.43               | 0.65 | High          |
| 6. Do you support destinations that are sustainably managed while preserving community identity and culture, and minimizing environmental impacts?                                                    | 4.26               | 0.69 | High          |
| 7. Are you willing to participate in green community activities or support eco-friendly accommodations, such as reforestation projects, plastic reduction, or carbon footprint reduction initiatives? | 4.45               | 0.63 | High          |
| Average                                                                                                                                                                                               | 4.39               | 0.66 | High          |

#### 4. Discussion

The findings of this study highlight the potential for developing low-carbon tourism (LCT) in the Ban Rak Thai community, which aligns with the principles of sustainable tourism and environmental responsibility. Low-carbon tourism emphasizes minimizing environmental impacts while simultaneously enhancing local economic benefits—objectives supported by existing research (Becken & Patterson, 2006; Gössling et al., 2018). These concepts underscore not only the importance of reducing emissions but also the importance of empowering communities and promoting long-term environmental stewardship.



**Figure 1** Prototype One-Day Low-Carbon Tourism Route in Ban Rak Thai Community, Mae Hong Son Province.

The greenhouse gas (GHG) emissions generated from each activity along the prototype routes are summarized in Table 5.

**Table 5** Greenhouse gas (GHG) emissions by activity (kg CO<sub>2</sub>eq/head).

| Activity                                              | GHG Emissions<br>(kg<br>CO <sub>2</sub> eq/head) | Reference                                                                    |
|-------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------|
| Tourism Activities                                    |                                                  |                                                                              |
| Stop 1: Tea Observation and Picking                   | No emission                                      | -                                                                            |
| Stop 2: Boat Tour, Reservoir Viewing, and Tea Tasting | 0.126                                            | Thai National LCI Database, TIIS-MTEC-NSTDA (with TGO electricity 2016-2018) |
| - Human-powered and solar-powered boat                |                                                  |                                                                              |
| - 10 passengers per trip                              |                                                  |                                                                              |
| - 100 g dried oolong tea per trip                     |                                                  |                                                                              |
| Stop 3: Nature Trail Hiking                           | No emission                                      | -                                                                            |
| Stop 4: Lunch (Single-dish meal)                      | 1.65                                             | Product category Rule of Tourism (PCR)                                       |
| Stop 5: Tea Brewing Workshop                          | 0.051                                            | Thai National LCI Database, TIIS-MTEC-NSTDA (with TGO electricity 2016-2018) |
| - Uses 10 g of organic dried oolong tea per person    |                                                  |                                                                              |
| Stop 6: Bicycle Tour (1.7 km)                         | No emission                                      | -                                                                            |
| Stop 7: Dinner (Thai-Chinese Buffet)                  | 2.27                                             | Product category Rule of Tourism (PCR)                                       |
| Transportation                                        |                                                  |                                                                              |
| Travel by Large Passenger Van (2000 cc, 10 seats)     | 2.746x49<br>= 134.53                             | - IPCC Vol. 2, Tables 3.2.1 and 3.2.2; DEDE                                  |
| - Diesel fuel, travel distance: 3.89 km               | = 13.453                                         | - Fuel efficiency data: Pollytion Control Department (2008)                  |
| - Fuel efficiency: 12.248 km/liter                    |                                                  |                                                                              |
| - Total fuel consumption: approximately 49 liters     |                                                  |                                                                              |

By applying the framework of sustainable development—with a focus on community participation and the conservation of natural resources—the analysis of Ban Rak Thai's tourism potential across eight dimensions, as proposed by Singsaktrakul and Sermkarndee (2013), strongly aligns with the core elements of ecotourism development. Notably, the dimensions of "tourism resources" and "environmentally friendly activities" emerged as significant contributors to Ban Rak Thai's capacity to become a preferred destination for environmentally conscious travelers, aligning with the strategic tourism development frameworks proposed by Crouch and Ritchie (1999).

The carbon footprint assessment of tourism activities revealed that most activities—such as hiking, cycling, and cultural learning—generated relatively low levels of greenhouse gas (GHG) emissions. These results align with the *IPCC Guidelines for National Greenhouse Gas Inventories* (IPCC, 2006) and Thailand's Product Category Rules (PCRs) for tourism activities, which emphasize the importance of quantifying and minimizing emissions across tourism value chains (Thaicarbonlabel, 2022).

In addition, the results support pro-environmental behavior theory, which posits that tourist behavior is influenced by awareness, beliefs, ethical values, and environmental motivations (Ajzen, 1991; Stern, 2000). The survey data revealed that tourists exhibited a high level of support for low-carbon tourism (*mean* = 4.35, *standard deviation* (*SD*) = 0.67), indicating that environmentally conscious practices and clear communication of sustainability measures can effectively attract like-minded visitors. This finding aligns with the research of Peeters and Dubois (2010), who reported a growing preference among European tourists for slow travel and rural experiences that prioritized local engagement and reduced travel emissions.

Moreover, the strategic planning process employed in this study, which integrated SWOT analysis with community participation, aligns with the principles of community-based tourism (CBT). CBT frameworks emphasize inclusive planning, in which local stakeholders play a central role in resource management and benefit distribution (Murphy, 1985; Simpson, 2008). Such participatory approaches are crucial for ensuring that tourism development remains context-specific and sustainable over the long term.

However, several limitations remain. These include challenges related to physical access to the village, continued dependence on fossil fuels, and inadequate waste management infrastructure. These findings align with those of Higham et al. (2016), who argue that the success of low-carbon tourism depends on investment in sustainable infrastructure and systematic support from government bodies.

## 5. Conclusions

This study confirms the feasibility and practical significance of implementing low-carbon tourism (LCT) models in Ban Rak Thai, Mae Hong Son Province, by leveraging local resources and enhancing community participation. The findings indicate that Ban Rak Thai holds significant tourism potential, characterized by rich natural landscapes, a distinctive Yunnanese-Chinese cultural heritage, and strong community-led services such as homestays and eco-conscious tourism practices. These internal strengths align with the principles of sustainable tourism and offer a solid foundation for LCT development. An analysis of

tourist behavior revealed a high level of interest and readiness among travelers to support environmentally friendly tourism initiatives, with strong preferences for green accommodations, clean transportation, and immersive local experiences that integrate carbon-reducing practices. This finding supports the theory of pro-environmental behavior, which suggests that tourists' ethical values and environmental awareness influence sustainable travel decisions. A carbon footprint assessment of two one-day prototype tourism routes developed through participatory planning with local stakeholders demonstrated that most tourism activities, including forest hiking, organic tea-making workshops, and cycling tours, produced minimal or no greenhouse gas emissions. Emissions were associated primarily with transportation and food consumption. These results validate the environmental potential of LCT in rural communities and offer data-driven insights into emission reduction strategies. Furthermore, the integration of SWOT analysis with community engagement proved effective in grounding tourism planning within a community-based tourism (CBT) framework. Localized strategic actions, such as promoting solar energy in homestays, developing slow-travel packages, and designing nature-based itineraries, help strengthen both environmental and economic resilience. However, key limitations were also identified, including limited access to renewable energy, insufficient waste management infrastructure, and a lack of low-carbon public transportation. These challenges emphasize the need for supportive policies, investment in sustainable infrastructure, and cross-sectoral collaboration to scale up LCT efforts and ensure long-term sustainability. Ultimately, Ban Rak Thai serves as a compelling model for rural low-carbon tourism development. By combining scientific carbon assessments, stakeholder engagement, and sustainability-focused planning, this study provides a replicable framework for designing tourism that minimizes environmental impact, empowers communities, and responds to the growing demand for climate-responsible travel.

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### Ethical considerations

This research was approved by the Human Research Ethics Committee of Thammasat University (Certificate No. 028/2566).

### Conflict of interest

The authors declare that they have no conflicts of interest.

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